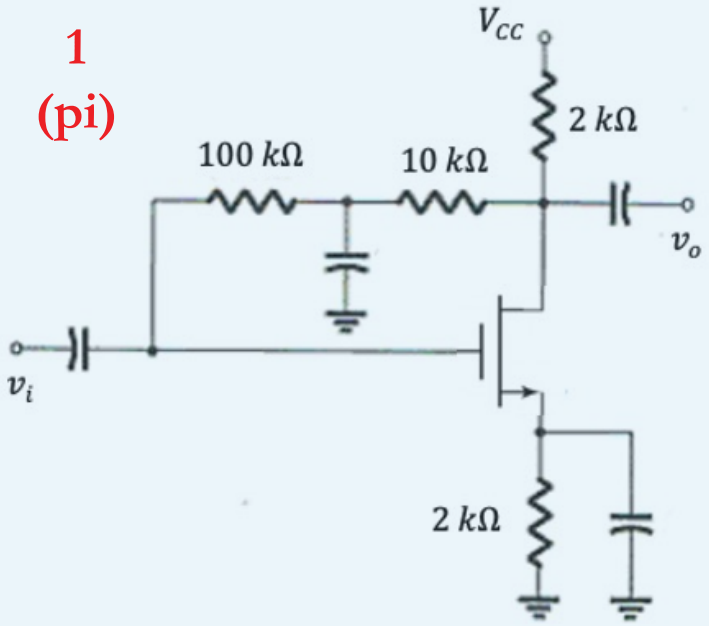
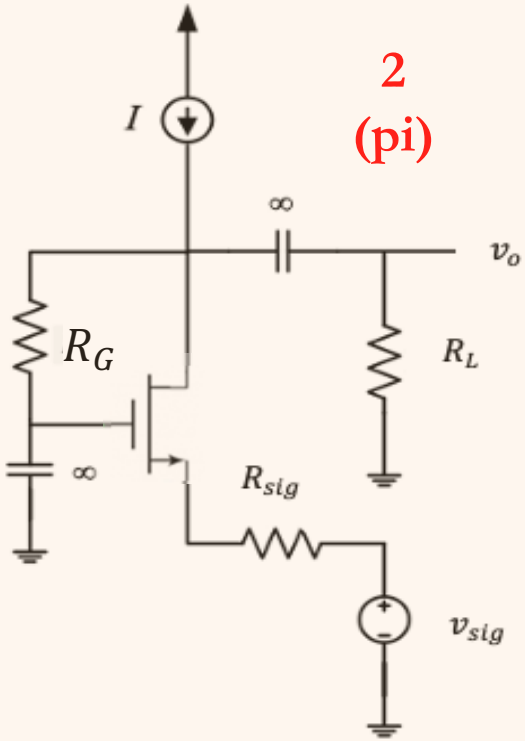
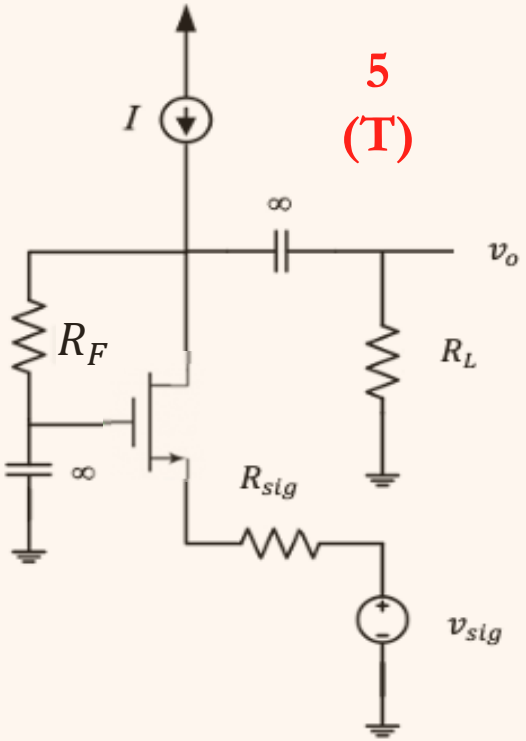
1
(pi)



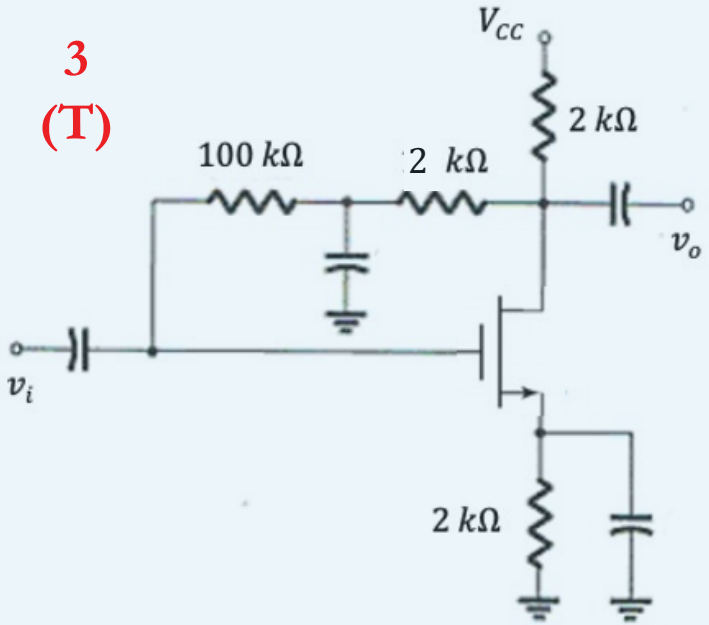
2
(pi)



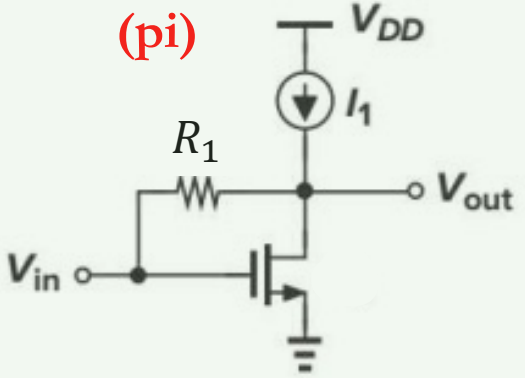
5
(T)



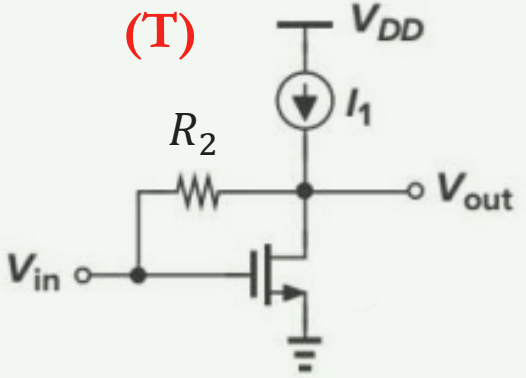
3
(T)



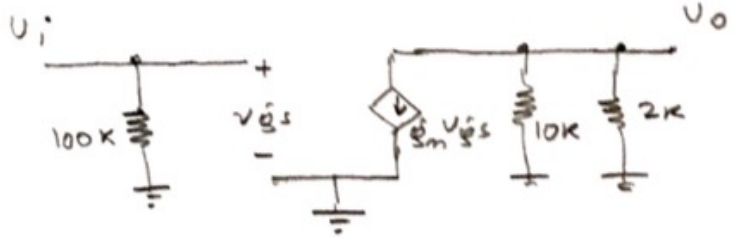
4
(pi)



6
(T)



①



$v_i = v_{gs}$

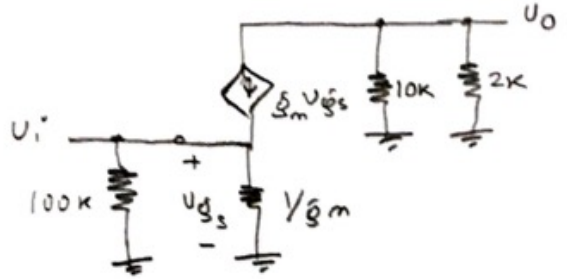
$v_o = -g_m v_{gs} (10 \| 2)k$

$A_v = \frac{v_o}{v_i} = -g_m (10K \| 2K)$

$R_i = 100K\ \Omega$

$R_o = (10K \| 2K)\ \Omega$

③



$v_i = -v_{gs}$

$v_o = -g_m v_{gs} (R_G \| R_L)$

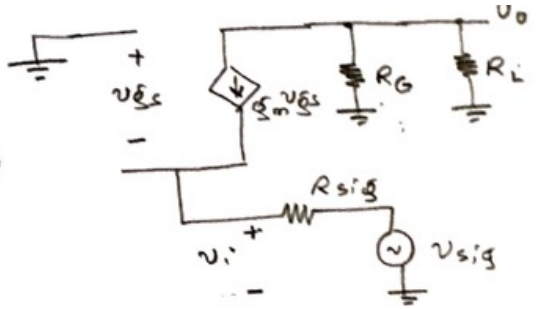
$A_v = \frac{v_o}{v_i} = g_m (R_G \| R_L)$

$i_i = \frac{-v_{gs}}{1/g_m} = \frac{v_i}{1/g_m}$

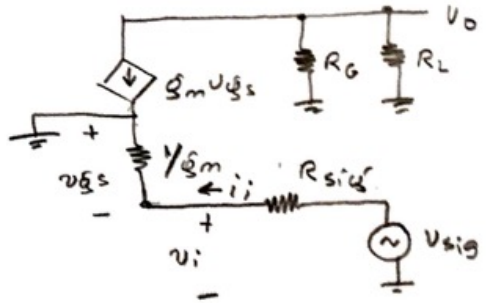
$R_i = \frac{v_i}{i_i} = 1/g_m$

$R_o = R_G$

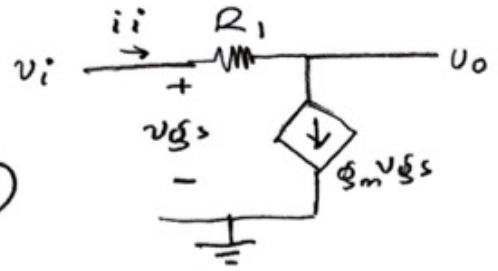
②



⑤



④



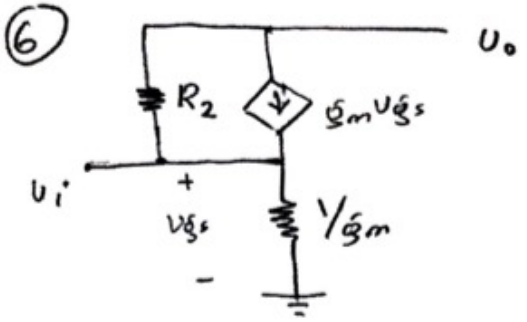
$$v_i = v_{gs}$$

$$\frac{v_i - v_o}{R_1} = g_m v_{gs} = g_m v_i$$

$$v_i (1 - g_m R_1) = v_o$$

$$A_v = \frac{v_o}{v_i} = 1 - g_m R_1$$

⑥



$$i_i = g_m v_i$$

$$R_i = 1/g_m$$

$$R_o = R_1$$